# ABINIT Q&A — LLM & RAG Evaluation

Topic Classification · Prompting Strategies · Agentic RAG vs. Native LLM

| | | | |
|---|---|---|---|
| **1,174** | **18** | **8** | **4.2/5** |
| gold Q&A pairs | topic categories | prompt conditions | best LLM score |

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Data: ABINIT mailing-list forum (3,866 topics) + Discourse forum (693 topics) ·
Judge: LLM-as-a-Judge (1–5 scale)

# We Asked the Expert.

*Xavier Gonze — ABINIT Project Director — has a message for all of us.*

> *When a user asks how to set ecut, you simply... tell them.*
>
> *...or you could let the AI do it.*
>
> *— Xavier Gonze (probably)*

**PRESS PLAY**

**What you are about to hear:**

- 🎙 Xavier's personal guide to answering ABINIT user questions
- 🤖 Generated entirely by AI (yes, really)
- 📊 Spoiler: the AI agrees with our results
- 😄 No directors were harmed in the making of this audio

# RAG vs. Native LLM — What, Why, and Why It Matters

Two approaches to AI-powered question answering · fundamentally different knowledge sources

## Native LLM

User question

**LLM (pre-trained weights)**
**Answers from memory**

Generated answer

- ✓ No external database needed
- ✓ Fast — single model call
- ✗ Limited to training data
- ✗ May hallucinate specific facts
- ✗ Cannot access recent updates

vs.

## Retrieval-Augmented Generation

User question

**Retrieve top-k docs**

**LLM + context**

Grounded answer (with sources)

- ✓ Answers grounded in real documents
- ✓ Can cite sources / verify facts
- ✓ Works on private / domain knowledge
- ✗ Quality depends on doc coverage
- ✗ Retrieval can miss relevant chunks

## Why It Matters for ABINIT

### Specific parameters

ecut, nstep, tolvrs,... exact values matter. LLMs may guess; RAG cites the manual.

### Version sensitivity

APIs change between ABINIT releases. RAG target the right doc.

### Hallucination risk

A wrong convergence or pseudopotential.

### Missing docs gap

15% of forum questions aren't in any doc. Docs need to improve.

### LLM from memory

Fast, broad, fluent — but may hallucinate code-specific details

### RAG from documents

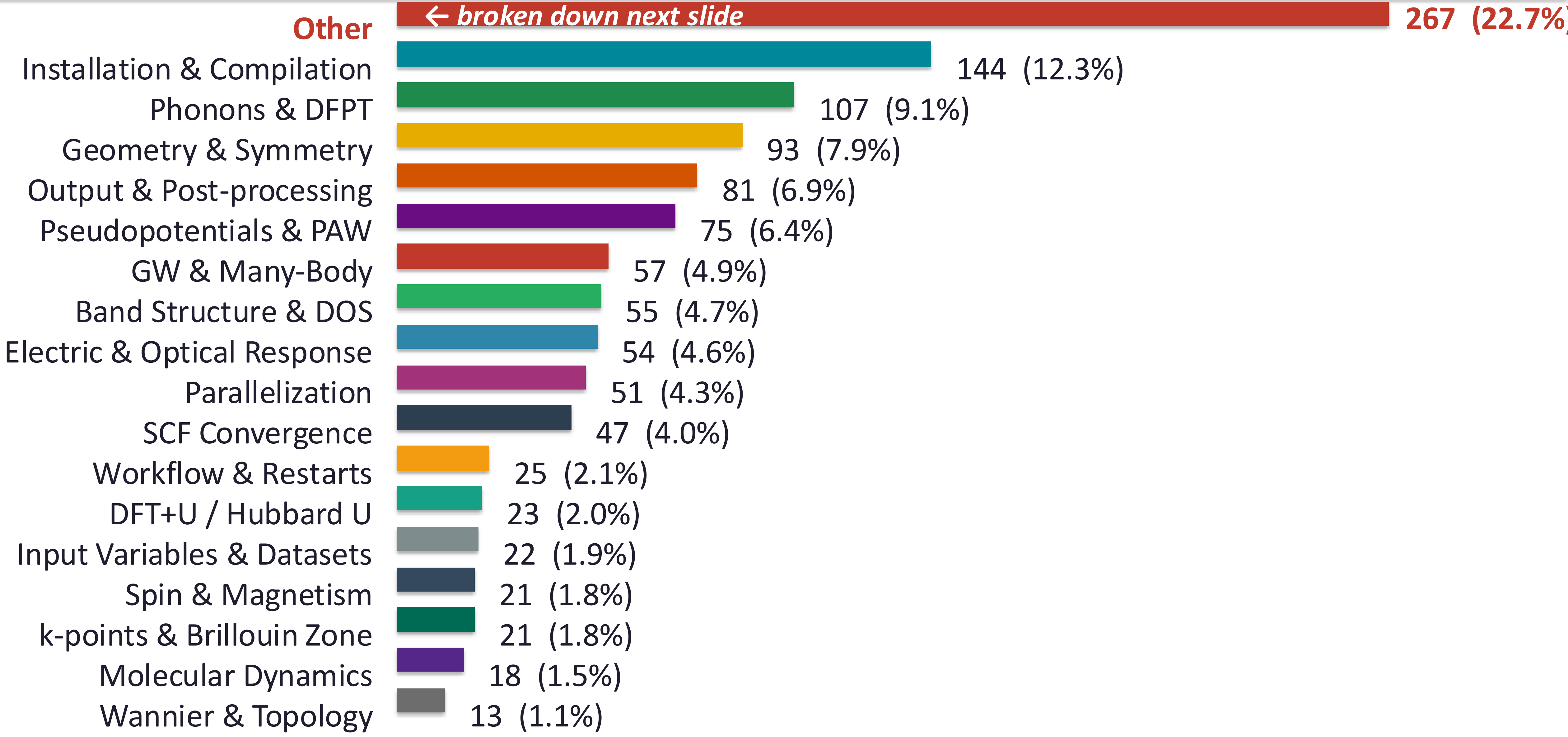
Grounded, citable — but only as good as the documentation

### Best of both worlds

LLM generates answer · RAG verifies facts · Users get reliable

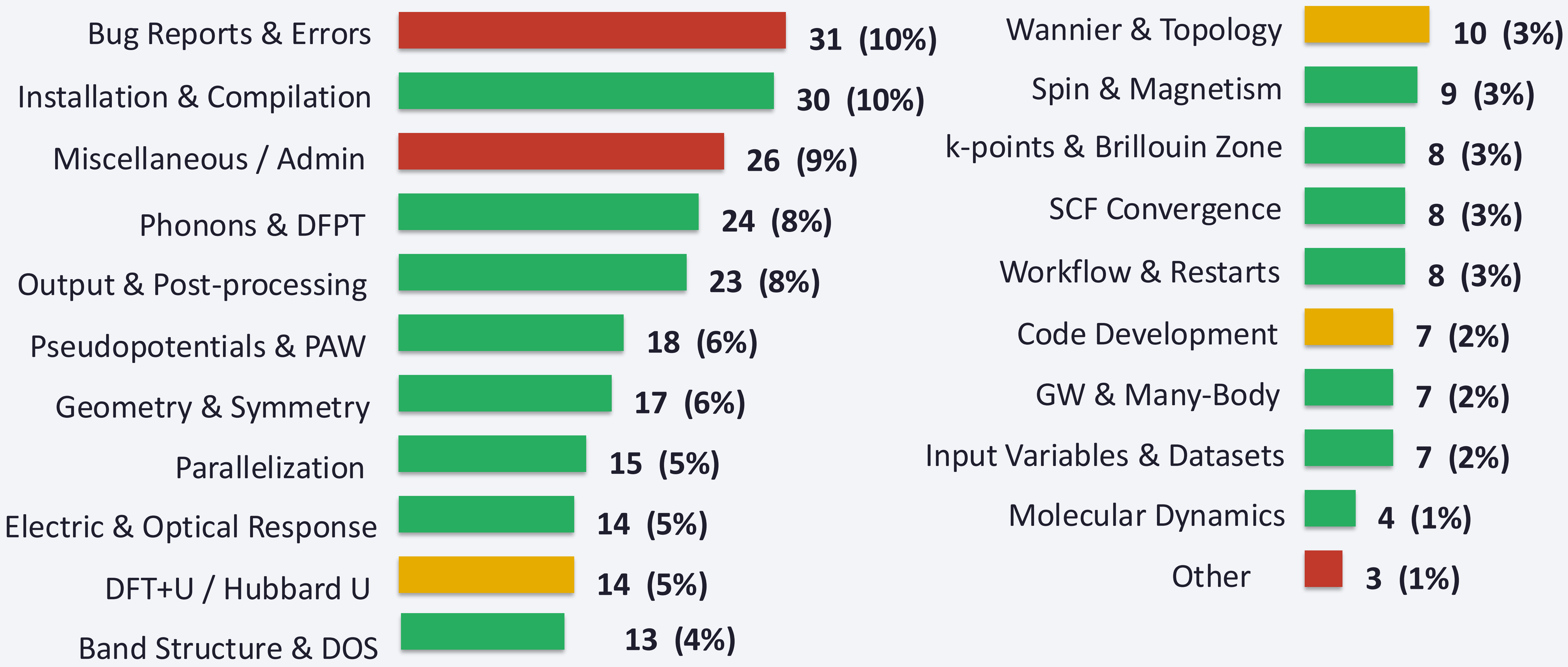
# Q&A Knowledge Base — 18 Topic Categories

1,174 gold Q&A pairs curated from ABINIT forums · classified by Claude Haiku · sorted by frequency



# Inside "Other" — 267 Pairs Sub-Classified

Only 9.7% truly miscellaneous · 60% classifier error · 13% missing topic categories




Classifier error — belongs to existing topic (205 pairs, ~60%)

Missing topic category (31 pairs, ~13%)

Truly misc: admin, bugs, dev (60 pairs, ~27%)

# Key Insights — RAG vs. Native LLM Performance

Score out of 5 · judge: gemini-3.1-flash-lite · 100–200 random forum questions

| System / Model | Configuration | Score / 5 |
|---|---|---|
| Agentic RAG (~200 docs) | Without reranker | 2.3 |
| Agentic RAG (~200 docs) | With reranker | 2.6 |
| Agentic RAG (~750 docs) | With reranker + Abipy context | 2.9 |
| Gemini CLI Agent (~750 docs) | find & grep on MD doc files | 2.7 |
| LLM — gemini-2.5-flash | 100 questions (June 2025) | 2.7 |
| LLM — gemma4_26b | 100 questions | 3.2 |
| LLM — haiku-3.5 | 100 questions | 3.3 |
| LLM — haiku-3.5 | 200 questions | 3.2 |
| LLM — gemini-3.1-flash-lite | 100 / 200 questions | 3.7 |
| **LLM — Sonnet 4.6** | 200 questions | 4.2 |
| **LLM — gemini-3.5-flash ★** | 100 questions (Top Performer) | 4.2 |

*★ Gemini-3.5-flash and Sonnet 4.6 both reach 4.2/5 · Modern LLMs have very likely seen the forum answers in their pre-training weights*

# RAG vs. LLM — Question-Level Breakdown

100 questions · comparing Agentic RAG vs. gemini-3.1-flash-lite · LLM judge

## 15 / 100 questions

### Both fail

Answers not aligned with expert.
Documentation may be outdated
or question has missing context.
Different approaches both valid.

## 4 / 100 questions

### RAG > LLM ✓

RAG finds the precise passage.
Highly specific input variables,
exact parameter values,
or version-specific fixes.

## 22 / 100 questions

### LLM > RAG

General DFT physics
(broad pre-training wins).
Incomplete user context.
System & build errors
not in documentation.

*Key finding: When both fail, documentation is often incomplete or question has insufficient context · RAG wins only when exact documentation exists*

# Why Does LLM Beat RAG? — Question Categories

Three categories of questions where documentation-based RAG cannot compete

## General DFT Physics

**Fundamental theory questions not tied to specific ABINIT syntax.**

These are answered better by broad LLM pre-training across thousands of DFT papers and textbooks. RAG is limited to ABINIT documentation which focuses on implementation, not theory.

**Forum examples:**

- *Is there a way to vary temperature in GS calculations?*
- *May I consider the structure has the same space group given negligible differences?*

## Incomplete Context

**User posts with implicit references not present in docs.**

User refers to an external error file, a specific geometry file, or cluster specs not mentioned in the question. RAG cannot retrieve relevant documents without fully-specified query context.

**Forum examples:**

- *I was wondering if this method is available in the latest ABINIT version.*
- *[References an attached file not included in the post]*

## System & Build Errors

**Compilation warnings, MPI failures, hardware issues.**

These errors depend on environment parameters (OS version, compiler, library paths) rarely resolved in ABINIT manual. LLMs have seen similar build errors across many HPC forums and Stack Overflow posts.

**Forum examples:**

- *Why does ABINIT 8.10.3 fail to compile with gfortran 9.2.1 on Ubuntu 18.04?*
- *Compile on MacBook with GCC + AVX processor*

# Impact of Missing Documentation

Curated subset of 73/100 questions (removing undocumented queries) · RAG improves significantly

## Full Dataset (100 questions)

**Agentic RAG**

2.6 / 5

**gemini-3.1-flash-lite**

3.7 / 5

Includes undocumented questions.
RAG limited by missing information.

## Curated Subset (73 questions — doc exists)

**Agentic RAG**

3.0 / 5

**gemini-3.1-flash-lite**

3.8 / 5

Removed questions where documentation lacked sufficient context.
RAG improves +0.4 pts.

## What This Tells Us

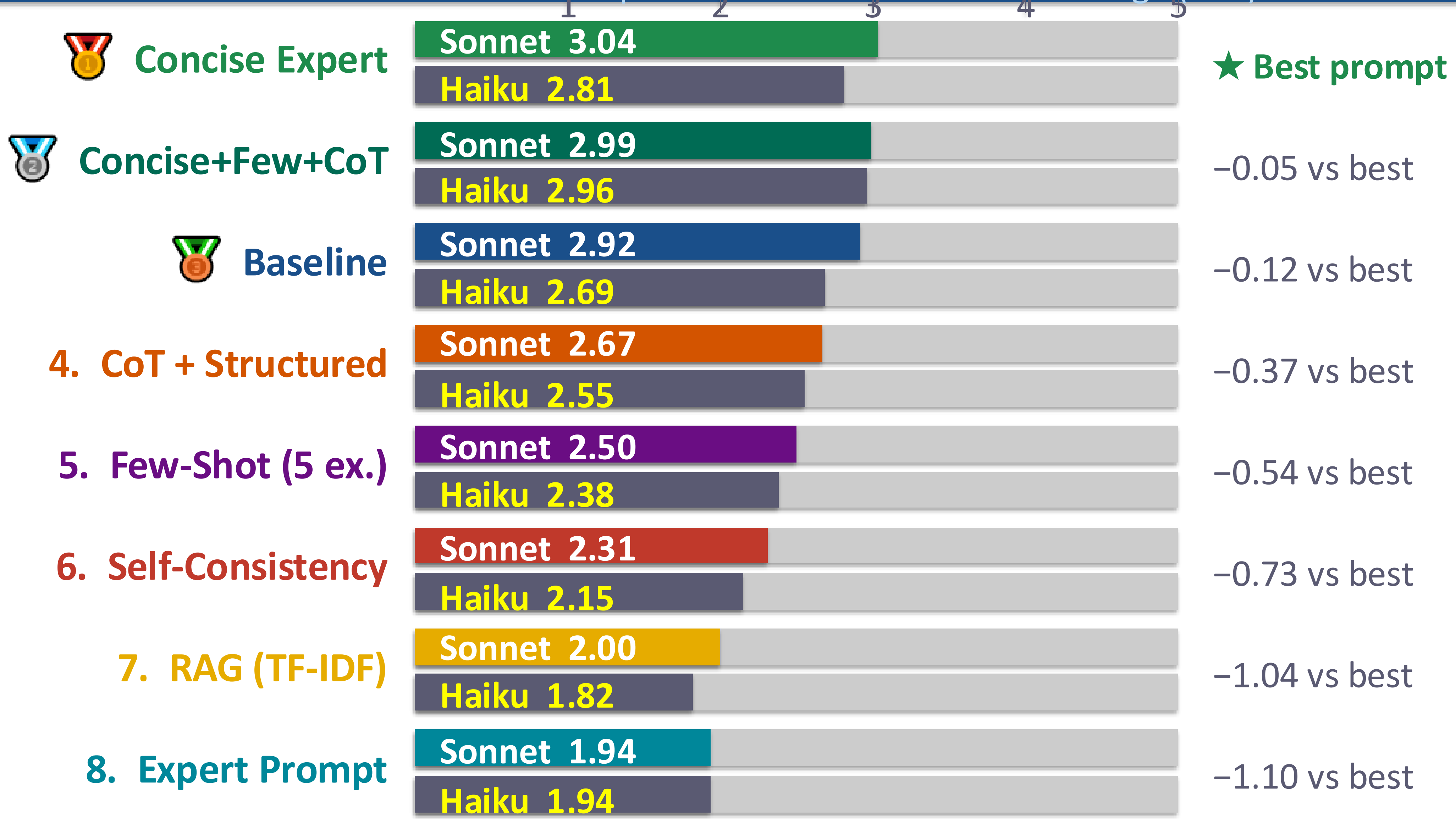
When the information actually exists in the documentation, RAG works well.

The retriever is not the problem, the documentation coverage is.

Improving documentation quality directly improves RAG performance.

# Prompting Strategy Evaluation — 8 Conditions

Claude Haiku 4.5 + Sonnet 4.6 · 200 forum questions · Structured Rubric Judge (1–5)



# Key Findings

Combined results: prompting strategies + Agentic RAG vs. Native LLM

### Modern LLMs already score >4/5 — pre-training includes forum content

Gemini-3.5-Flash and Sonnet 4.6 both reach 4.2/5 against human expert answers. These models have almost certainly seen the ABINIT forum in their training data. This is a high bar for any RAG system to beat without much better documentation coverage.

### Prompt style is decisive — Concise Expert reaches 3.04/5 (structured judge)

Concise Expert outperforms all 7 other prompting conditions. Short, professionally-toned instructions beat verbose expert personas by +57%. Adding CoT or few-shot examples gives marginal gains once the base prompt is well-tuned.

### RAG is effective — when the documentation exists

Removing undocumented questions from the evaluation set improves Agentic RAG by +0.4 pts. The retriever works well when the information is present. Documentation coverage, not retrieval quality, is the primary bottleneck.

# Key Findings

Combined results: prompting strategies + Agentic RAG vs. Native LLM

### TF-IDF RAG underperforms — semantic retrieval is essential

Our TF-IDF RAG scored only 2.0/5 on the structured judge, below the plain Concise Expert (3.04/5). Agentic RAG with dense embeddings + reranker reaches 2.9/5. Hybrid search (BM25 + embeddings) is the recommended approach.

### Questions where both fail reveal documentation gaps

15/100 questions where both LLM and RAG fail indicate missing, outdated, or ambiguous documentation. These become a prioritized documentation improvement list — a systematic way to identify where ABINIT docs need strengthening.

# How Can We Use This? — Practical Applications

Turning evaluation results into actionable improvements for the ABINIT community

### Publish Advised Prompts

Curate and publish a community-maintained library of optimized prompts for ABINIT (phonons, GW, parallelization...). Users paste them into any LLM for instantly better answers.

### Documentation Gap Detection

Questions where both LLM and RAG score < 3 = missing or outdated documentation. Run this evaluation quarterly on new forum posts → auto-generate a prioritized list of documentation topics that need improvement.

### RAG as LLM Fact-Checker

Don't replace LLM with RAG — use RAG to validate LLM output. LLM generates a fluent answer; RAG verifies specific input variable names, allowed values, and version constraints against the official manual. Flags hallucinations.

### Tiered Answer System

Route questions by type: General DFT physics → pure LLM (scores > 4/5). Specific ABINIT variables → LLM + RAG validation. Build/install questions → redirect to known-working forum threads. Reduces hallucination risk.

### Forum-Powered Knowledge Base

Continuously auto-score new forum posts (the scoring pipeline already exists). Gold-quality answers (score ≥ 4) are automatically added to the RAG corpus. The knowledge base grows with the community. Documentation becomes a living resource.

### ABINIT LLM Benchmark Suite

Test new ABINIT releases: if LLM + RAG scores drop, documentation regressed. Compare Claude, Gemini, open-source models systematically. Enables data-driven decisions about which AI tools to recommend to users.

# Conclusions

## Agentic RAG vs. Native LLM

### LLMs score >4/5

Modern LLMs already surpass Agentic RAG. Pre-training on forum threads gives them a major advantage.

### RAG works when docs exist

Removing undocumented questions: RAG +0.4 pts. The retriever is not the bottleneck — coverage is.

### RAG as validation layer

Best use: fact-check LLM output on specific variables and values, not as primary answer generator.

### BERTScore confirms it

All top models cluster 0.81–0.83 F1. RAG leads on Recall; LLMs lead on Precision.

## Prompting Strategy Evaluation

### Best: 'Concise Expert' → 3.04/5

Concise, professional tone beats verbose expert personas. Style alignment matters as much as content.

### CoT helps Sonnet more than Haiku

Larger models benefit more from reasoning scaffolds. Chain-of-thought adds ~+0.1 pts for Sonnet.

### Combining strategies: marginal gain

Concise+Few+CoT reaches 2.99/5 — nearly identical to plain Concise Expert (3.04). More is not always better.

### Self-consistency disappoints

Selecting among 5 imperfect answers doesn't help when none contains the specific forum resolution.

# BERTScore Validation — Semantic Similarity

roberta-large · F1 / Precision / Recall vs. human expert answers · confirms LLM judge ranking

BERTScore computes similarity using contextual embeddings — captures semantic equivalence, not just keyword overlap. "The car is fast" ≈ "The vehicle is quick". All top models cluster tightly (0.81–0.83) confirming the LLM judge ranking is robust.

| Model / System | F1 | Precision | Recall |
|---|---|---|---|
| **RAG Agentic RAG (Reranker, ~750 docs)** | **0.8280** | **0.8081** | **0.8495** |
| Gemini-3.1-Flash-Lite | 0.8226 | 0.7945 | 0.8531 |
| Gemini-3.5-Flash | 0.8198 | 0.7913 | 0.8508 |
| Gemma4 26B | 0.8194 | 0.7940 | 0.8468 |
| Haiku 3.5 (100 Qs) | 0.8176 | 0.7900 | 0.8475 |
| Haiku 3.5 (200 Qs) | 0.8164 | 0.7880 | 0.8472 |
| Sonnet 4.6 (200 Qs) | 0.8112 | 0.7788 | 0.8466 |

*Note: All models cluster in 0.81–0.83 F1 range · RAG leads on Recall (finds relevant content) · LLMs lead on Precision (less hallucination drift)*